

Mid-term exam, Machine Learning (MDS), April 3rd, 2024

- You have 1h to solve the exam
- All questions have equal weight
- Please justify all your answers

Question 1: Given the example of categorical Naive Bayes from the theory slides:

<i>Outlook</i>	<i>Temperature</i>	<i>Humidity</i>	<i>Wind</i>	<i>Play ball</i>
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

- a) How many parameters uses Naive Bayes for this problem? And with Laplace smoothing?
- b) What steps would you take in order to use Gaussian Naive Bayes for this problem? Do you think it would make sense? Would it need more or less parameters than categorical NB?

Question 2: In an univariate regression problem, find the equation for the best horizontal line that fits a set of input training points using least squares. You can assume your input data is $\{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\}$ where each $(x_i, y_i) \in \mathbb{R}^2$.

Question 3: You have access to a set of input finite (labelled) data. Explain in your own words how to:

- a) Estimate a model's generalization performance as accurately as possible
- b) Choose among competing models
- c) Estimate the parameters of a model